

$$\Omega = \{\tau_1, \dots, \tau_K\}$$

$$\mathbf{w} = (w_1, \dots, w_K)^\top$$

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \|\mathbf{w} - \mathbf{f}\|_2^2$$

$$\text{s.t.} \quad \sum_{i=1}^K w_i = 1, \quad w_i \geq 0$$